Pricing 2025 claim cost in a developed and an emerging market

Team AIIAA (Yichuan Lin, Oussama El Hadi)
IAA AI-Assisted Hackathon 2026, 30 October 2026

We priced motor, home buildings and home contents for a developed market (Part 1) and an emerging market (Part 2). The ranked Part 1 entry, S09, scored 0.13689 on the private board (rank 12 of 51); the team's final Part 1 method, a blend with a teammate's pipeline, scored 0.13718 as a late submission, which is not ranked. The Part 2 candidate P2-S15 scores 0.15807 on the public board (rank 18 of 38 on 30 October), against 0.14370 for our first emerging-market file.

1 Pricing performance and adaptation (40%)

1.1 Task, data and target

Each 2025 policy row needs the ultimate cost of claims with a loss date in the row's period, net of excess and recoveries and excluding declined claims, as a row total. Scoring is Tweedie D^2 with power 1.5. Part 1 has 2,409,311 training rows for 2021 to 2024 and 163,150 claims valued at 31 December 2024; Part 2 has two training years, 2023 and 2024 (Table 1).

	Developed	Emerging		Developed	Emerging
Test rows	626,639	209,063	New business in 2025 test rows	15%	76%
Motor exposure-years	399,468	38,969	Test policies seen in training	85%	24%
Motor claim frequency	13.48%	18.09%	Broker share, motor exposure	25%	70%
Motor mean severity (CU)	3,862	2,250	Missing values, where affected	2.5%	12%

Table 1: Developed against emerging market. Motor figures are for 2023-24, severity is reported cost, and the missing-value shares are approximate.

A claim attaches to the row of its policy whose period contains the loss date, at incurred minus recoveries. Repudiated claims are excluded, and the 410 Part 1 claims that match no row are listed and left out; totals reconcile with a difference of 0 (C005). The claims file is one snapshot, so instead of a triangle we fit a cost-weighted reporting-lag curve per line on accident year 2021 and divide each row's reported cost by its completeness (C006). Motor 2024 pure premium rises from 454 reported to 585 developed, inside the 564 to 606 range of 2021 to 2023. Bodily injury is 43% of motor cost and only 14% of it is reported at 30 days. Claims are capped at the line's 99.5th percentile and a per-line load adds the excess back (C014). Ten data issues, such as exposure stored in days and sums insured inflated by 1,000, are handled in code at load time, and the raw files are never edited (C007, Appendix A).

1.2 Part 1 model

All models fit capped cost per unit of completeness-weighted exposure with a Tweedie loss of power 1.5 (C010): a banded GLM, a spline GAM, LightGBM, XGBoost and CatBoost. Each is rescaled to its training total, because log-link LightGBM ended at 0.83 of actual cost on its own training rows (C013), and a feature policy drops columns the trees overfitted, such as motor `make_model` (C026). Per-line blend weights from out-of-fold predictions are used only if they beat equal weights on both folds (C021, C030). Half the fitted trend takes predictions to 2025 (motor 1.018, buildings 1.095, contents 1.064; C019), and the final refit averages three seeds (C029). That is S09. Our teammate's independent pipeline (OBLEMOS: a Poisson by Gamma GLM with BIC selection, a P-spline GAM, LightGBM and CatBoost, averaged as Blend4) correlates 0.965 with S09. After the close we put Blend4 on S09's level per line and blended $0.7 \times \text{S09} + 0.3 \times \text{Blend4}$; this late submission is the team's final Part 1 method.

1.3 Adapting to the emerging market

In the emerging market 76% of the 2025 rows are new business, against 8% of its training rows (C033), and the construction codes, occupation text and age bands differ. The same code therefore

runs with a market setting and a shared schema (C034, C035). The adaptation took six steps, and each change was kept only if it passed the rule in force at the time (Section 3).

1. Emerging data alone (P2-S01). With about 500 claims per fit year on each property line, the GLM and GAM overfit (buildings time-fold `d2_cal` of −10.5 and −30.7).
2. Pooled training (C038, C040). Developed rows join every fit at weight w times exposure, with `market` as a feature, and the level is re-balanced on the emerging fit rows. Time-fold `d2_cal` rose from 0.0762 to 0.0915 on motor, 0.0410 to 0.1065 on buildings and 0.0779 to 0.1072 on contents; validation set $w = 1$ for buildings and 0.3 for the other lines.
3. Level (C043). Buildings and contents keep the developed half trend (1.0948 and 1.0636), because the 2023 fire year (buildings 2024 over 2023 is 0.63) rules out their own. Motor pure premium rose 10.7% in 2024 against 1.75% applied, and six public-board probes (P2-S06 to P2-S11) moved motor to its own full trend, 1.1606. This is our one decision taken mainly from the public board.
4. New business (C044, C047). The Part 2 audit found validation rows to be 92% renewals, against 77% new business in the test (P2-F1), so views re-weighted to the test mix joined the gate. Contents became 0.35× the pooled-0.3 GLM + 0.65× the pooled-1 LightGBM (P2-S12). The 68,823 motor new-business rows take an equal 1/12 blend of GLM, LightGBM, XGBoost and CatBoost from three training sets (pooled at weights 0.3 and 1, and new-business rows only), because searched weights lost the guard (P2-S13). Motor renewal rows keep the four pooled-0.3 models at 0.25 each.
5. Teammate's GAM (C054). Refitted on the pooled data, the OBLEMOS Part 1 GAM beat the pooled CatBoost on the buildings time fold (0.1257 against 0.1176). Buildings became 0.4× that GAM + 0.6× the pooled-1 CatBoost, at the P2-S13 buildings total (P2-S14).
6. Seed averaging (C057). P2-S15 averages the LightGBM, XGBoost and CatBoost test predictions over three seeds, as S09 did, and keeps the P2-S13 buildings total (C058). Out-of-fold predictions do not change, so no fold gate applies; the rule, fixed before the score was seen, was to keep it unless it fell more than 0.0005 below P2-S14.

Of the 0.0144 public gain from P2-S01 to P2-S15, the developed half trend gave 0.0009, pooling 0.0073, the buildings weight and motor level 0.0022 and the four steps of Table 2 0.0039 (Table 3).

	Raw		Test mix		New business only	
Adopted step	time	grouped	time	grouped	time	grouped
P2-S12, contents blend	+0.0005	+0.0032	+0.0020	+0.0041	n/r	+0.0039
P2-S13, motor new-business rows	+0.0001	+0.0009	+0.0008	+0.0050	+0.0012	+0.0032
P2-S14, buildings with the GAM	+0.0068	+0.0018	+0.0052	+0.0058	+0.0048	+0.0068
P2-S15, three-seed refit	out-of-fold predictions unchanged					

Table 2: Gain in `d2_cal` on the six Part 2 validation views over the recipe each step replaced, on that line: P2-S05 contents, P2-S12 motor and P2-S13 buildings (C044, C047, C054). The P2-S12 grouped gains were measured on the folds its weights were searched on; its time-fold gains are out of search. n/r: not reported.

1.4 Assumptions and limitations

Both parts assume that open-claim case estimates are unbiased and that the accident year 2021 reporting pattern holds for later years and for the emerging market; Part 1 also assumes that the construction codes T and W both mean timber. The emerging market gives one time fold with about 500 claims a year on each property line, its reporting curves are borrowed, and nothing beyond 24 months can be checked: the Part 2 audit puts the resulting overstatement of the 2024 ultimate at 1.8% for motor and 2.5% for buildings (P2-F9). New-business evidence is 37 to 697 claims per line and scheme, enough to reject a change but not to tune one. Part of the 2025 level rests on public-board probes, the motor trend (+0.0029) and the half trend (+0.0009), and Part 1 showed

	File	Recipe	Public	Private	Rank
Part 1	S09	Five models, guarded weights, half trend	0.12863	0.13689	12 of 51
	Blend4	Teammate's pipeline alone	0.12736	0.13526	not ranked
	S09 blend	$0.7\times$S09 $+\,0.3\times$Blend4, final Part 1 method	0.12949	0.13718	late, not ranked
Part 2	baseline	Constant average cost	0.07319		
	P2-S01	Emerging data only	0.14370		8 of 10 (24 Oct)
	P2-S04	Pooled training, $w = 0.3$ on all lines	0.15193		9 of 12 (25 Oct)
	P2-S10	Buildings at $w = 1$, motor at its own trend	0.15414		
	P2-S13	Contents blend, motor new-business segment	0.15577		23 of 35 (29 Oct)
	P2-S14	Teammate's GAM in buildings	0.15781		
	P2-S15	P2-S14 with three-seed refit (final candidate)	0.15807		18 of 38 (30 Oct)

Table 3: Kaggle Tweedie D^2. The Part 1 rank is the final private rank; Part 2 ranks are public-board team ranks on the date shown, and Part 2 private scores follow the close on 31 October. On 30 October the public median was 0.15773 and the top score 0.17556.

that such a read can flip sign on the private board. The Part 2 private score is unknown. Appendix F lists the remaining limitations.

1.5 Office premium for the 20 reference policies

R is the P2-S15 prediction for the row, already a row total. From the emerging column of `expenses.parquet`, catastrophe reinsurance is Cat $= 0.12\,R$ on buildings and contents and 0 on motor (its unit is a share of property risk premium, and we count contents as property). The fixed expense $F = 85$ CU is charged in full per policy. The share-of-premium items add to $k = 0.15$ (variable expense) + commission (0.20 broker, 0 direct) + 0.08 (premium tax) + 0.08 (target profit); none of the 20 policies uses the comparison-site channel, whose value is blank. There is no minimum or maximum premium.

$$P = \frac{R + \text{Cat} + F}{1 - k}, \qquad 1 - k = 0.49 \text{ (broker)}, \quad 0.69 \text{ (direct)}.$$

For B00229821, $387.92 + 46.55 + 85.00 = 519.47$ and $519.47/0.49 = 1{,}060.14$. Table 4 gives all 20 policies and Appendix D splits each P by item.

row_id	R	Cat	P	WP	row_id	R	Cat	P	WP
M00118569$^{\text{d,r}}$	922.76	0.00	1,460.52	1,095.48	B00218282	391.04	46.92	1,067.27	230.80
M00129413	212.69	0.00	607.53	354.15	B00223369	515.20	61.82	1,351.08	204.56
M00153851	368.16	0.00	924.81	414.72	B00229821	387.92	46.55	1,060.14	285.60
M00161473	1,135.24	0.00	2,490.28	1,131.78	B00250679	38.71	4.65	261.95	180.72
M00167496	777.40	0.00	1,760.00	463.95	C00259568^{r}	40.88	4.91	266.92	83.18
M00176777	1,072.30	0.00	2,361.85	443.61	C00270331	37.45	4.49	259.07	140.29
M00185311^{d}	1,105.96	0.00	1,726.03	872.31	C00283470	86.59	10.39	371.39	117.97
B00215707	342.12	41.05	955.46	272.44	C00286287	162.14	19.46	544.08	269.23
B00215809^{d}	480.49	57.66	903.11	237.22	C00302985	60.81	7.30	312.46	115.88
B00217584	825.49	99.06	2,060.29	323.36	C00314425	5.97	0.72	187.11	102.87

Table 4: Office premium build-up in CU, with $F = 85.00$ on every row. Row ids starting M, B and C are motor, buildings and contents. All rows are broker new business ($1 - k = 0.49$) except d: direct ($1 - k = 0.69$) and r: renewal. WP is the written premium under the current tariff. Source: `scripts/part2_office_premium.py --predictions work/p2_s15_seeds3.csv` (C060).

The 20 rows sum to $R = 8{,}969.31$ and $P = 20{,}931.34$, against 7,340.12 of written premium, which is the current tariff and not a target for our 2025 cost. P exceeds written premium on every row, by 1.33 to 6.60 times (median 2.92), mostly through the loadings: a broker row with $R = 0$ would still cost 173.47, and F, Cat and the shares make up 36% to 97% of P. R alone exceeds written premium on 10 rows, 6 buildings and 4 motor, all new business; our buildings new-business predictions ran 1.30 times actual on the grouped folds (77 claims), so part of that gap may be our own overstatement.

Written premium does not scale with exposure, so F dominates the three short rows. Pro-rating F by exposure would lower P from 607.53 to 491.56 for M00129413 (exposure 0.33), from 261.95 to 107.49 for B00250679 (0.11) and from 187.11 to 25.52 for C00314425 (0.07). These conventions are sign-off item R15.

2 Effective use of AI (25%)

2.1 How the AI worked

We used Claude Code in three roles. A coordinator (claude-fable-5-1) planned each step, wrote the decision rule before a run and proposed what to submit. Workers (claude-opus-5-5 and claude-sonnet-5-5) wrote a failing test, then the code, ran the experiment and logged the change in `docs/CHANGELOG.md`. Read-only auditors trained nothing and backed each finding with a script: 17 scripts behind B1 to B8 in Part 1 and 27 checks behind P2-F1 to P2-F10 in Part 2. `docs/AI_LOG.md` (A01 to A41) records for every step what the AI produced, how it was checked and whether it was accepted, modified or rejected. Our teammate ran his Part 2 experiments in his own Claude session (C048 to C056, A27).

The AI worked in every stage. It profiled all 11 files against the sample submission (A02), counted each data issue with a script (A03), scanned every column for leakage (A06) and wrote features and models test-first (A07). In the first full run it traced an XGBoost crash to a boolean column with missing values, and a 9 to 18% under-prediction of the trees to the loss of balance in log-link Tweedie boosting, testing each cause before changing code (A08). Its vectorised weight search covers 888,030 weight vectors for eight candidates, and the threaded version cut the motor search from 1,277 to 306 seconds with the same weights (C041).

2.2 Where a check corrected or stopped the AI

AI output reached a submission only after a check. Table 5 lists proposals that failed one; the checks came from other agents, from tests and from the data.

AI proposal	Check	Outcome
Per-peril reporting curves (coordinator, C022)	Worker compared them with the pooled curve: largest difference 5e-16	No effect; kept as an input (A09)
GLM market interactions in a 9-model blend, +0.0024, above the rule (C042)	Worse alone on all six contents folds; new-business level up 21% with no test of it	Rejected by the coordinator (A23)
Buildings renewal segment that passed the gate (C047)	Its gain came from a new-business-only GLM applied to renewal rows	Rejected (A26)
Motor control run, better on every view (C047)	It was run after the result was seen	Diagnostic only, not submitted (A26)
Ridge GLM in the transfer model (C048)	Shrinkage test: a penalty of 1e9 must return the base model	Scaling bug fixed before any result (A27)
Note that a teammate's Part 1 file scored 0.1356 in public	Kaggle listing with labelled columns: 0.13562 was its private score	Note corrected; S09 led on both boards (A28)

Table 5: AI proposals that a check changed or stopped (`docs/AI_LOG.md`, `docs/CHANGELOG.md`).

2.3 What the humans decided

People made the judgement calls. The team lead approved items R01 to R11 of the sign-off sheet (cleaning rules, development, cap, trend, feature drops, Part 2 schema, borrowed curves and pooled design) on 25 October in a single reply without item-level reasons, and the sheet records that. R12 to R16 (the motor level, the test-mix gate, the new-business segment, the office-premium conventions and P2-S15) were signed the same way on 30 October, so all 16 are signed. The team asked for XGBoost and the GAM (A07, C011), built the OBLEMOS pipeline, chose the 0.7/0.3 blend as the Part 1 method and P2-S15 as the Part 2 final with P2-S14 as the hedge. After the misread score in Table 5, the Part 2 selection compared scores from one board, with the six views, on files the repository rebuilds.

3 Trust, validation and guardrails (20%)

3.1 Validation design

No split is random and the year is not a feature. Part 1 uses two expanding-window folds that score 2023 and 2024. Part 2 has a time fold (train 2023, score 2024) and a grouped 5-fold split on `policy_id`, both scored on emerging rows (C036). Our validation score `d2_cal` rescales each fold to its actual total, so it measures ranking, and trees early-stop on a random 10% of the fitting rows, never on the scored fold (C027). Each Part 2 candidate is scored on six views: the time fold and the grouped mean, each raw, re-weighted so that new business carries its share of 2025 test exposure (about 0.77) and on new-business rows only (C044). The re-weighted view tracks the board: the P2-S03 to P2-S04 step was +0.0154 raw, +0.0051 re-weighted and +0.0050 in public. Weights are still searched on raw folds: 37 to 697 new-business claims can reject a change but not tune one.

3.2 Adoption rules

Rules were fixed before the runs they judged. A Part 1 change needed a mean gain of at least 0.002 with no fold lost (C018). Searched blend weights must beat equal weights on every scheme, enforced in code (C030). A Part 2 change must not lose on the raw time fold, the raw grouped mean or the test-mix grouped mean, and must gain 0.002 on the raw mean or the test-mix grouped mean (C044). After a contents candidate with negative time-fold test-mix and new-business views lost 0.00017 in public, we also ruled out any loss on those two views (C053). Gaps under 0.0005 are ties, won by the simpler option. Among the Part 2 blend candidates tried in public, only those that gained on all six views raised the score.

3.3 Data, code and reproduction

A SHA256 manifest, read-only file attributes and a test guard every raw byte (C007), and the developed files in the Part 2 release matched the Part 1 manifest (C033). The suite has 342 pytest cases in 24 files, and modules are written test-first. The blend step rebuilds S09 byte for byte from the stored model predictions (same SHA256 as the upload; a full refit differs slightly between runs, as the tree refits are not bit-reproducible), and P2-S14, rebuilt by two routes, differs from the upload by at most 5.1e-10 (C058). P2-S15 keeps the row ids, row order and buildings total of P2-S14 (correlation 0.9991).

3.4 Audits and human sign-off

The Part 1 audit's findings on `make_model`, `branch_id` and early stopping became C026 and C027; its trend double-counting finding (B4) was probed and not adopted. The Part 2 audit changed our validation (P2-F1) and caught a diagnostic that read uncorrected exposure (P2-F8). `docs/HUMAN_REVIEW.md` lists 16 decisions (R01 to R16) with their evidence by file and line.

4 Communication and documentation (15%)

`README.md` gives the task, layout, pinned environment, commands, pipeline, governance and known limitations. `docs/CHANGELOG.md` has 62 entries (C000 to C061), each with its reason, evidence and code location, and code comments cite the same ids. `docs/SUBMISSIONS.md` lists every upload with its recipe, validation and scores. `docs/AI_LOG.md`, `docs/HUMAN_REVIEW.md` and the two audit files hold the AI record, `docs/references/` archives the Kaggle pages, and `docs/part2_validation/` keeps the Part 2 fold scores. The evidence board in `docs/dashboard/` puts the main figures on one page, and the five-minute video follows `docs/VIDEO_SCRIPT.md`. The sign-off sheet and team notes are in Chinese; code and logs are in English.

A reviewer rebuilds each file as follows. `reproduce.ps1` runs from the raw data to S09 (the blend step from stored predictions matches the upload's SHA256; a full refit differs slightly between runs) and with `-Part2` to P2-S13. `docs/PART2_OBLEMOS.md` gives the two commands for P2-S14, which `blend_runs.py --write-from` with `--level-from` also builds (C058). P2-S15 is that write step after a three-seed refit (`oof_predictions.py --refit-only --seeds 3`), read with `--pred-tag s3` (C057, C059). The Part 1 blend comes from `oblemos_part1/blend_with_S09.py`.

Appendix, not part of the five graded pages

A Data-quality issues and treatment

Issue	Size	Treatment
Q1 `exposure` holds day counts or a 360-day basis (1.0139 = 365/360)	5,723 motor, 4,176 buildings, 4,305 contents rows	Recomputed from period dates on a 365-day year (C001)
Q2 motor `customer_id` lost leading zeros	98,110 training and 25,554 test rows	Padded to 8 characters; 100% join to customers (C002)
Q3 `sum_insured` inflated by 1,000	Values above 5m, including 370 buildings and 360 contents test rows	Divided by 1,000 and flagged; the same policies differ by exactly 1,000 across years (C003)
Q4 `construction_type` case and codes T/W	13,250 policies switch from T to W	Lower-cased; T and W mapped to timber (C004)
Q5 claims matching no policy row	294 motor, 67 buildings, 49 contents	Excluded from the target and listed; reconciliation difference 0 (C005)
Q6 `customer_service_contacts`	About 3.0 on claimed rows against 0.41 on unclaimed rows	Dropped as leakage (C008)
Q7 about 2.5% random missing values	Many columns	Not imputed for tree models; the GAM uses median imputation plus missing indicators (C011)
Q8 `telematics_score` empty for 87% of motor rows	Structural: no device	Not imputed
Q9 `renewal_decision` only in training	All rows	Dropped; known only after the row ends
Q10 training and test drift	Median `renewal_count` 6 to 3, `vehicle_age` 7 to 9, `ncb_years` 7 to 8	Year is not a feature; the audit found the drift internally consistent (`renewal_count` +1.0 per year)

Table 6: Data-quality issues in the developed market and their treatment (`docs/DATA_GUIDE.md`, `docs/CHANGELOG.md`).

Part 2 re-check (C033): Q1 recurs in the emerging training tables only (253 motor rows); Q2 is absent; Q3 recurs and the 5m threshold still separates; Q4 changes to the codes BRK, STN, WOD and MIX (C035); 36 claims match no row; missing values rise to about 12%; the Q8 columns are absent. `customer_service_contacts` again separates claimed rows (3.4 against 0.42 contacts) and stays out.

B All submissions

ID	Recipe	Public	Private
S01	Equal blend of GLM, GAM, LightGBM, XGBoost; full trend	0.12453	0.13347
S02	S01 with no trend	0.12376	0.13466
S03	S01 with half trend	0.12464	0.13456
S04	Tuned LightGBM and XGBoost, equal weights, half trend	0.12477	0.13279
S05	Equal blend of GLM, GAM and tuned trees	0.12464	0.13477
S06	Per-line weights from out-of-fold predictions (C021)	0.12579	0.13505
S07	Feature policy, inner early stopping, CatBoost, guarded weights	0.12851	0.13693
S08	S07 with a lower trend on buildings and contents (probe)	0.12820	0.13703
S09	S07 recipe with seed-averaged refit (ranked final, rank 12 of 51)	0.12863	0.13689
Blend4	Teammate's pipeline alone (not ranked)	0.12736	0.13526
Late	$0.5\times$S09 $+\,0.5\times$Blend4	0.12971	0.13707
Late	$0.7\times$S09 $+\,0.3\times$Blend4 (final Part 1 method)	0.12949	0.13718

Table 7: Part 1 submissions. Private scores come from the Kaggle listing after the close. Late submissions are not ranked; the scored 0.7/0.3 file was the teammate's upload, and the repository holds a copy rebuilt by `oblemos_part1/blend_with_S09.py`.

ID	Recipe	Public
P2-S01	Emerging data only, no trend	0.14370
P2-S02	P2-S01 with developed half trend	0.14460
P2-S03	Contents pooled	0.14693
P2-S04	All lines pooled, $w = 0.3$	0.15193
P2-S05	P2-S04 with buildings at $w = 1.0$	0.15123
P2-S06	Level probe, all lines $\times 1.10$	0.15206
P2-S07	Level probe, motor $\times 1.10$	0.15362
P2-S08	Level probe, buildings $\times 1.10$	0.15029
P2-S09	Motor at the square root of its own trend, 1.0773	0.15284
P2-S10	Motor at its full own trend, 1.1606	0.15414
P2-S11	P2-S10 uplift on motor new business only (tie, not adopted)	0.15434
P2-S12	P2-S10 with the contents blend across weight runs	0.15474
P2-S13	P2-S12 with the motor new-business segment blend	0.15577
P2-S14	P2-S13 with buildings = $0.4\times$ teammate GAM + $0.6\times$ pooled CatBoost, at the P2-S13 buildings total	0.15781
P2-S15	P2-S14 with three-seed averaged LightGBM, XGBoost and CatBoost (final candidate; hedge P2-S14)	0.15807

Table 8: Part 2 submissions, public scores (`docs/SUBMISSIONS.md`). Five further probes uploaded by our teammate on P2-S12 and P2-S13 scored 0.15483 to 0.15563 and none was adopted.

C Part 2 validation detail

Line	Emerging only	Pooled, $w = 0.3$	Pooled, $w = 1.0$	Adopted w
Motor	0.0762 / 0.0677	0.0915 / 0.0740	0.0908 / 0.0730	0.3
Buildings	0.0410 / 0.0732	0.1065 / 0.1132	0.1169 / 0.1180	1.0
Contents	0.0779 / 0.0934	0.1072 / 0.1100	0.1069 / 0.1128	0.3

Table 9: Emerging-only against pooled training, `d2_cal` (time fold / grouped mean).

The GAM left the pooled runs after it exceeded the memory cap on 756k rows. Tree models needed level re-balancing on emerging rows, with factors of 1.08 to 1.26 at weight 0.3 on the property lines and 1.01 to 1.10 at weight 1.0. P2-S05 (buildings at $w = 1.0$) scored 0.15123 in public against 0.15193 for P2-S04, so the board disagreed with validation; we kept the validation rule (C040).

Recipe	Time	Grouped	Grouped test-mix	Public
P2-S01	0.10971	0.12306	0.13006	0.14370
P2-S03	0.11546	0.12623	0.13421	0.14693
P2-S04	0.14179	0.14165	0.13934	0.15193
P2-S05	0.14486	0.14311	0.13973	0.15123

Table 10: Audit evidence for P2-F1: Kaggle-like D^2 of the three lines stacked, each line sum-calibrated.

Line	Time fold, new / renewal	Grouped, new / renewal	New-business claims, time / grouped
Motor	1.012 / 0.933	0.812 / 1.025	382 / 697
Buildings	2.229 / 1.607	1.296 / 0.981	37 / 77
Contents	0.737 / 1.111	0.837 / 1.018	65 / 102

Table 11: Predicted-to-actual on reported cost for the P2-S05 blend, by segment. Source: `docs/part2_new_business_check.md`.

Line	Days	Emerging 2023	Developed 2023, same truncation	Developed 2021, used
Motor	30	0.603	0.579	0.556
Motor	90	0.774	0.717	0.723
Motor	365	0.961	0.970	0.922
Buildings	30	0.913	0.829	0.857
Buildings	90	0.952	0.927	0.933
Buildings	365	0.998	0.992	0.992
Contents	30	0.961	0.947	0.949
Contents	90	0.991	0.995	0.989
Contents	365	0.993	0.996	0.990

Table 12: Cost-weighted cumulative share reported (`docs/part2_emerging_target_summary.txt`, C037).

The peril mix differs most on buildings: fire is 0.454 of emerging cost against 0.237 in developed accident year 2021. Mean completeness of the 2024 motor rows is 0.7672 with the developed peril shares and 0.8202 with the emerging ones.

D Office premium split by item

row_id	R + Cat + F	Variable expense	Commission	Premium tax	Profit	P
M00118569	1,007.76	219.08	0.00	116.84	116.84	1,460.52
M00129413	297.69	91.13	121.51	48.60	48.60	607.53
M00153851	453.16	138.72	184.96	73.99	73.99	924.81
M00161473	1,220.24	373.54	498.06	199.22	199.22	2,490.28
M00167496	862.40	264.00	352.00	140.80	140.80	1,760.00
M00176777	1,157.30	354.28	472.37	188.95	188.95	2,361.85
M00185311	1,190.96	258.90	0.00	138.08	138.08	1,726.03
B00215707	468.17	143.32	191.09	76.44	76.44	955.46
B00215809	623.14	135.47	0.00	72.25	72.25	903.11
B00217584	1,009.54	309.04	412.06	164.82	164.82	2,060.29
B00218282	522.96	160.09	213.45	85.38	85.38	1,067.27
B00223369	662.03	202.66	270.22	108.09	108.09	1,351.08
B00229821	519.47	159.02	212.03	84.81	84.81	1,060.14
B00250679	128.36	39.29	52.39	20.96	20.96	261.95
C00259568	130.79	40.04	53.38	21.35	21.35	266.92
C00270331	126.94	38.86	51.81	20.73	20.73	259.07
C00283470	181.98	55.71	74.28	29.71	29.71	371.39
C00286287	266.60	81.61	108.82	43.53	43.53	544.08
C00302985	153.10	46.87	62.49	25.00	25.00	312.46
C00314425	91.68	28.07	37.42	14.97	14.97	187.11

Table 13: Each share-of-premium item in CU; every row adds up to P. Exact exposures and the full table come from `scripts/part2_office_premium.py --predictions work/p2_s15_seeds3.csv`.

E Part 1 target and validation detail

Repudiated claims (1,711, all with zero amounts) are excluded (C005), and 20.7% of 2024 motor claims were still open at 31 December 2024. The per-line caps are 51,585, 147,445 and 28,449 CU for motor, buildings and contents in Part 1 and 27,578, 220,808 and 28,312 CU in the emerging market (C014, C037); the capped share of cost is 8.8%, 4.2% and 2.1% in Part 1 and 5.0%, 4.9% and 2.1% in the emerging market. Motor perils other than bodily injury are 76% to 99% reported at 30 days. Rescaling each model to its training total raised motor validation D^2 from 0.066 to 0.073 (C013). Per-line tuning was adopted only where the mean gain was at least 0.002: motor LightGBM gained 0.0075, motor XGBoost 0.0031 and contents XGBoost 0.0024 (C015, C018). The feature policy (C026) raised mean LightGBM `d2_cal` from 0.0832 to 0.0887 on motor, from 0.1094 to 0.1134 on buildings and from 0.0891 to 0.0936 on contents. Motor searched weights lost to equal weights on

the 2024 fold (0.09031 against 0.09071), so motor uses equal weights (C021, C030). The S09 blends scored a mean two-fold `d2_cal` of 0.0919 for motor, 0.1144 for buildings and 0.0945 for contents.

Rejected Part 1 alternatives: per-peril frequency by severity scored 0.0801 against 0.0893 on motor, 0.1045 against 0.1094 on buildings and 0.0879 against 0.0911 on contents, with 2024 bodily-injury counts over-predicted by 1.43 (C023). An out-of-fold target encoding of `make_model` carried weak signal (code relativities correlated 0.18 across years, against −0.05 for shuffled codes), and the equal blend gained 0.0003 on the mean but fell on 2024 (C031). Re-tuning the trees on the final feature set gave motor LightGBM +0.0027 on the mean but −0.0008 on 2024, so nothing was adopted (C032).

F Assumptions and limitations

Both parts assume that open-claim case estimates are unbiased and that the accident year 2021 reporting pattern holds for later years and for the emerging market. Part 1 also assumes that the construction codes T and W both mean timber (C004).

The emerging market's two years give one time fold, with about 500 claims per year on each property line, and 2024 is also the least developed year. The reporting curves are borrowed, nothing beyond 24 months can be checked, and 20% of 2024 emerging motor claims are open. Buildings 2023 was a fire year (49 fire claims for 3.22m CU against 21 for 1.13m in 2024). The large-loss load is uniform within a line, and the emerging caps rest on 36, 6 and 6 claims above the cap.

Part of the level rests on public-board probes: the developed half trend (P2-S02, +0.0009, below our 0.002 rule) and the motor level. The motor gain (+0.0029) is larger, and the model trained on 2023 had also under-predicted 2024 motor by 6%. The risk is that a level read from about a third of the rows does not hold on the rest. In Part 1 the S08 probe of audit finding B4 flipped sign between the boards (public −0.0003, private +0.0001).

Part 1 audit items left open: the trend may count the year level twice (B4; residual trend 1.017, 1.048 and 1.004 against 1.018, 1.095 and 1.064 applied). The 2025 motor large-loss load includes under-reported 2024 bodily injury and may be 1.4% low (B5). Recoveries appear only on closed claims, so open 2024 own-damage cost is about 1.4% overstated (B6). Prior-claim history is developed in cost but not in claim counts (B7). Open-claim reserving for bodily injury cannot be verified from one snapshot, and the customer snapshot date is unknown, so early training rows may see later information. Buildings and contents cost above the cap concentrates in the top risk decile while the load is uniform, and the GAM uses a single penalty (C011). In the motor new-business segment, a control run made after the result was seen scored higher on every view; choosing it would have been a post-hoc selection, so it stays a diagnostic.